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Olympiad Diophantine equations

Level 4 Olympiad · Number theory

A proof-based toolkit for integer equations: modular obstructions, p-adic valuations, size arguments, descent and exponential equations.

By the end you can: You will be able to choose a modulus that rules out solutions and justify it completely.; You will be able to use valuations, Legendre's formula and squeezing between powers to bound solutions.; You will be able to write full BMO-standard solutions to exponential and descent problems..

Lesson

Show that $x^2 + y^2 = 3z^2$ has no solutions in positive integers. Try a few values and nothing works, but trying values proves nothing. Instead, look modulo 3: squares are 0 or 1, so $x^2 + y^2 \equiv 0 \pmod 3$ forces $x^2 \equiv y^2 \equiv 0$, that is, $3 \mid x$ and $3 \mid y$. Then $9 \mid 3z^2$, so $3 \mid z^2$ and $3 \mid z$. Dividing everything by 3 gives a new solution $(x/3, y/3, z/3)$ with a smaller z. If there were any positive solution, there would be one with z as small as possible, and we have just produced a smaller one. Contradiction.

That four-line proof used two tools: a modulus in which squares are restricted, and descent. Olympiad Diophantine equations are rarely solved by one clever trick. They are solved by a small toolkit used with judgement: modular obstructions, p-adic valuations, size arguments, descent and factorisation. This unit builds that toolkit and, as importantly, trains you to choose the right tool quickly.

Modular obstructions

A modulus is useful when the powers in the equation take very few values. The ones to know by heart:

- squares: $\{0, 1\}$ modulo 3 and 4; $\{0, 1, 4\}$ modulo 8; $\{0, \pm 1\}$ modulo 5;
- cubes: $\{0, \pm 1\}$ modulo 7 and modulo 9;
- fourth powers: $\{0, 1\}$ modulo 5 and modulo 16.

Why these moduli? If $k \mid p-1$ and $p \nmid x$, then $(x^k)^{(p-1)/k} = x^{p-1} \equiv 1$, so x^k is a root of $y^{(p-1)/k} \equiv 1$, which by Lagrange's Theorem has at most $(p-1)/k$ roots. So k-th powers take at most $1 + \frac{p-1}{k}$ values modulo p. Against cubes, use 7 or 13; against fifth powers, 11; against sixth powers, 7 or 13. Powers of a fixed base, such as 2^n or 3^n, are periodic with period equal to the order, so 3^n takes only the values $1, 3, 9$ modulo 13.

Example 1. Find all positive integers n and m with $n! + 3 = m^2$.

Solution.
Try small cases: $n = 1$ gives $4 = 2^2$, $n = 2$ gives 5, $n = 3$ gives $9 = 3^2$, $n = 4$ gives 27, $n = 5$ gives 123, and $n = 6$ gives 723, which lies between $26^2 = 676$ and $27^2 = 729$. So we expect only $n = 1$ and $n = 3$.
For large n, $n!$ is divisible by every small number, so modulo any fixed M the left side is eventually just 3. We therefore want a modulus in which 3 is not a square. Modulo 3 is the obvious first try, but it fails, since $3 \equiv 0$ is a square. Modulo 4 works: 3 is not a square modulo 4, and $4 \mid n!$ for all $n \geq 4$. So for $n \geq 4$ we would have $m^2 \equiv 3 \pmod 4$, which is impossible.
The solutions are $(n, m) = (1, 2)$ and $(3, 3)$. The moral: when a factorial or a high power

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appears, it vanishes modulo many numbers, and you should pick the modulus in which what is left over is not a square.

p-adic valuations

For a prime p and a nonzero integer n, $v_p(n)$ is the exponent of p in n. The rules are $v_p(ab) = v_p(a) + v_p(b)$ and $v_p(a+b) \geq \min(v_p(a), v_p(b))$, with equality when $v_p(a) \neq v_p(b)$ (factor out the smaller power of p; one term is left divisible by p and the other is not). A positive integer is a perfect k-th power if and only if k divides every one of its valuations.

Coprime products. If a, b are coprime positive integers and ab is a perfect k-th power, then a and b are both perfect k-th powers.

Legendre's Formula. $v_p(n!) = \displaystyle\sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p-1}$, where $s_p(n)$ is the sum of the digits of n in base p.

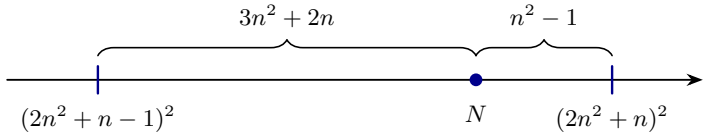
Proofs. For coprime products: each prime divides at most one of a, b, so $v_p(a)$ is either 0 or $v_p(ab)$, a multiple of k in both cases. For Legendre: $v_p(n!) = \sum_{m \leq n} v_p(m)$, and $v_p(m)$ counts the $i \geq 1$ with $p^i \mid m$. Swapping the order of counting, for each i there are $\lfloor n/p^i \rfloor$ multiples of p^i up to n. For the digit form, if $n = \sum_j d_j p^j$ then $\lfloor n/p^i \rfloor = \sum_{j \geq i} d_j p^{j-i}$, and summing over $i \geq 1$ gives $\sum_j d_j \frac{p^j - 1}{p-1} = \frac{n - s_p(n)}{p-1}$. □

Example 2. For which integers $n \geq 2$ is $n!$ divisible by 2^{n-2}?

Solution.
The direct approach is to estimate $\lfloor n/2 \rfloor + \lfloor n/4 \rfloor + \cdots$, which is a little less than n, and then fight with the error terms from the floors. The key observation is that the base 2 form of Legendre's Formula makes the error exact: $v_2(n!) = n - s_2(n)$, where $s_2(n)$ is the number of 1s in the binary expansion of n.
So we need $n - s_2(n) \geq n - 2$, that is, $s_2(n) \leq 2$. The answer is: n is a power of 2, or a sum of two distinct powers of 2. As a check, $3 = 2 + 1$ works since $2 \mid 6$, while $7 = 111_2$ fails: $v_2(7!) = 4 < 5$.

Size and bounding

Integers cannot be squeezed. If $k^2 < N < (k+1)^2$ for some integer k, then N is not a square, so to show that an expression N is never a square, trap it between two consecutive squares. To find the right squares, take the square root of the leading terms. For $N = 4n^4 + 4n^3 + 1$, the top terms suggest $\sqrt{N} \approx 2n^2 + n$, and indeed $(2n^2+n)^2 = 4n^4 + 4n^3 + n^2$. Comparing, $(2n^2+n)^2 - N = n^2 - 1$ and $N - (2n^2 + n - 1)^2 = 3n^2 + 2n$, both positive for $n \geq 2$. So among positive integers, N is a square only for $n = 1$, when it equals 9.



$N = 4n^4 + 4n^3 + 1$: for $n \geq 2$ both gaps are positive, so no square fits.

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Other size arguments to keep in mind: order the variables ($x \le y \le z$) and see which term dominates; factorials and exponentials outgrow polynomials; and in a factorised equation, each factor divides a fixed number, which leaves finitely many cases.

Example 3. Find all positive integers x and y with $x^2 + 3^x = y^2$.

Solution.
With x in an exponent, it is tempting to start reducing modulo things. But $y^2 - x^2$ is crying out to be factorised: $(y - x)(y + x) = 3^x$. Since the right side is positive and $y + x > 0$, we have $y > x$, so both factors are positive, and hence both are powers of 3. Write $y - x = 3^i$ and $y + x = 3^{x-i}$, where $i < x - i$ because $y - x < y + x$.
Subtracting, $2x = 3^{x-i} - 3^i = 3^i(3^{x-2i} - 1)$. Put $j = x - 2i \ge 1$, so that $x = 2i + j$ and

$$2(2i + j) = 3^i(3^j - 1).$$

Now the key observation: the left side is linear in i and j, the right side is exponential, so only small values can work.
We use some easy inequalities, each true at its first value and preserved at every step, because the left side at least triples while the right side only grows by a constant: $3^j - 1 \ge 2j$ for $j \ge 1$ (and likewise $3^j - 1 \ge 4j$ for $j \ge 2$), and $3^i \ge 4i + 1$ for $i \ge 2$. If $i \ge 2$, the right side is at least $(4i + 1) \cdot 2j = 8ij + 2j > 4i + 2j$, so there are no solutions. If $i = 1$, we need $4 + 2j = 3(3^j - 1)$; $j = 1$ works ($6 = 6$), and for $j \ge 2$ we have $3^j - 1 \ge 4j$, so the right side is at least $12j > 2j + 4$.
If $i = 0$, we need $2j = 3^j - 1$; $j = 1$ works and $j \ge 2$ gives $3^j - 1 \ge 4j > 2j$.
The case $(i, j) = (0, 1)$ gives $x = 1$, $y = 2$, and $(i, j) = (1, 1)$ gives $x = 3$, $y = 6$. Both check: $1 + 3 = 4$ and $9 + 27 = 36$. So the solutions are $(1, 2)$ and $(3, 6)$.

Infinite descent

Descent is induction run backwards: if every solution produces a smaller positive solution, there are no solutions at all, because positive integers cannot decrease for ever. Write it as a minimal counterexample: "take a solution with $|z|$ as small as possible", then produce a smaller one. The usual engine is a homogeneous equation in which some prime is forced to divide every variable. For example, in $x^3 + 2y^3 = 4z^3$, x must be even; writing $x = 2x'$ and dividing by 2 gives $4x'^3 + y^3 = 2z^3$, so y is even, and then z is even, and $(x/2, y/2, z/2)$ is again a solution. So the only solution is $x = y = z = 0$.

Exponential equations

Equations such as $3^a - 2^b = 1$ have three standard moves. Factorise, typically as a difference of squares when an exponent is even, and control the gcd of the factors. Reduce modulo a power of one base to pin down the other exponent modulo an order. Count the powers of a prime on each side, by parity arguments or LTE.

Example 4. Find all non-negative integers a and b with $7^a = 3 \cdot 2^b + 1$.

Solution.
Small cases give $7 = 3 \cdot 2 + 1$ and $49 = 3 \cdot 16 + 1$, so $(a, b) = (1, 1)$ and $(2, 4)$. Clearly $a \ge 1$.
Rewrite as $7^a - 1 = 3 \cdot 2^b$: the left side must have no prime factors except 2 and 3, and that is very restrictive. The parity of a decides which move to make.
If a is odd, then $7^a - 1 = 6(7^{a-1} + 7^{a-2} + \cdots + 1)$, and the bracket is a sum of a odd numbers, so it is odd. Hence $v_2(7^a - 1) = 1$, so $b = 1$ and $7^a = 7$, giving $a = 1$.

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If $a = 2c$ is even, factorise: $(7^c - 1)(7^c + 1) = 3 \cdot 2^b$. Since $7 \equiv 1 \pmod 3$, the factor $7^c - 1$ is divisible by 3, so $7^c + 1 \equiv 2 \pmod 3$ is not. Being a divisor of $3 \cdot 2^b$ that is coprime to 3, $7^c + 1$ is a power of 2. If c is even, $7^c \equiv 1 \pmod 4$, so $7^c + 1 \equiv 2 \pmod 4$ and $7^c + 1 = 2$, which is impossible for $c \geq 1$. If c is odd, $7^c + 1 = 8(7^{c-1} - 7^{c-2} + \cdots + 1)$ with an odd bracket (again c odd terms), so $7^c + 1 = 8$ and $c = 1$. Then $a = 2$ and $48 = 3 \cdot 2^b$ gives $b = 4$.
So the solutions are $(a, b) = (1, 1)$ and $(2, 4)$.

Common traps.

- "Integers" includes zero and negative numbers. Read the question and handle them.
- Finding solutions by testing is not a proof that there are no others. You must prove a bound or an obstruction.
- When you factorise, check the signs of the factors and their gcd before splitting into cases.
- The coprime products lemma needs coprimality. If the factors share a factor of 2, as in Example 4, deal with it first.
- A modular obstruction is only a proof if you show the complete list of residues.
- In descent, check that the new solution is smaller and still a genuine solution; homogeneous equations always have the zero solution, which must be excluded or listed.

How to spot this in a contest

If testing finds a few small solutions, the problem is about proving there are no large ones: think factorisation (each factor divides something fixed) or squeezing between consecutive powers. If testing finds nothing, look for a modulus, chosen by the powers present (squares: 4, 8, 3, 5; cubes: 7, 9; fixed bases: their orders), or for descent if the equation is homogeneous. Factorials suggest valuations and Legendre's Formula. A product equal to a power suggests valuations and the coprime products lemma. Two exponential bases suggest working modulo a power of one base, then using orders, LTE or a difference of squares.

At BMO standard, a full solution states the answer, checks that the claimed solutions work, and then proves that there are no others. Markers look for every case to be closed, so make your case split explicit.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ Prove that there are no integers x and y with $x^3 + y^3 = 2026$.

2. ⋆⋆ Find all positive integers n for which $3^n + 55$ is a perfect square.

3. ⋆⋆ When 2026! is written in base 12, how many zeros does it end with?

4. ⋆⋆⋆ Prove that the only integer solution of $x^2 + y^2 + z^2 = 7w^2$ is $x = y = z = w = 0$.

5. ⋆⋆⋆ Find all positive integers n for which $n^4 + 2n^3 + 6n + 7$ is a perfect square.

6. ⋆⋆⋆⋆ Find all pairs of positive integers (a, b) such that $7^a - 3^b = 4$.

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Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AIME 1987 Problem 5 ⋆
2. UKMT BMO1 2001/02 Problem 1 ⋆
3. UKMT BMO1 2016/17 Problem 3 ⋆⋆
4. USAMO 1979 Problem 1 ⋆⋆
5. AIME 1989 Problem 9 ⋆⋆
6. UKMT BMO1 2012/13 Problem 4 ⋆⋆
7. UKMT BMO1 2011/12 Problem 5 ⋆⋆⋆
8. USAMO 1976 Problem 3 ⋆⋆⋆
9. USAJMO 2011 Problem 1 ⋆⋆⋆
10. UKMT BMO1 2018/19 Problem 6 ⋆⋆⋆
11. USAJMO 2013 Problem 1 ⋆⋆⋆
12. UKMT BMO1 2006/07 Problem 6 ⋆⋆⋆⋆

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Hints

1. Cubes take very few values modulo 9 and modulo 7. Try both: only one of them works.
2. First decide the parity of n by working modulo 4. Then factorise.
3. $12 = 2^2 \cdot 3$. Compute $v_2(2026!)$ and $v_3(2026!)$, and do not assume which prime is the bottleneck.
4. Squares modulo 8 are 0, 1 or 4. Split on the parity of w, and descend when w is even.
5. Compare with $(n^2+n)^2$ and $(n^2+n-1)^2$, and factorise the difference with the first.
6. If $b \geq 2$, work modulo 9 to find a modulo 3. Then look for a prime modulo which 3 has order 3.

Solutions

Problem 1

Modulo 9 is the natural first try, but it does not help: cubes are 0 or ± 1 modulo 9, so $x^3 + y^3$ can be any of $0, \pm 1, \pm 2$, and $2026 \equiv 1 \pmod 9$.

Work modulo 7 instead. If $7 \nmid x$ then $x^6 \equiv 1$ by Fermat's Little Theorem, so $x^3 \equiv \pm 1$; hence every cube is congruent to 0, 1 or 6 modulo 7. The possible values of $x^3 + y^3$ modulo 7 are therefore $0+0=0$, $0+1=1$, $0+6=6$, $1+1=2$, $1+6 \equiv 0$ and $6+6 \equiv 5$, that is, $0, 1, 2, 5, 6$. But $2026 = 7 \cdot 289 + 3 \equiv 3 \pmod 7$. So the equation has no integer solutions. □

Problem 2

Answer: $n = 2$ and $n = 6$

If n is odd then $3^n \equiv (-1)^n = -1 \pmod 4$, so $3^n + 55 \equiv 54 \equiv 2 \pmod 4$, which is not a square. So $n = 2k$ is even.

Write $3^{2k} + 55 = m^2$ with $m > 0$. Then $(m - 3^k)(m + 3^k) = 55$. The second factor is positive, so the first is positive too, and it is the smaller of the two. The factorisations of 55 into two positive factors are $1 \cdot 55$ and $5 \cdot 11$. Subtracting the factors gives $2 \cdot 3^k = 54$ or $2 \cdot 3^k = 6$, so $k = 3$ or $k = 1$.

Thus $n = 6$, with $729 + 55 = 784 = 28^2$, and $n = 2$, with $9 + 55 = 64 = 8^2$. These are the only solutions.

Problem 3

Answer: 1009

The number of trailing zeros in base 12 is the largest k with $12^k \mid 2026!$. Since $12^k = 2^{2k}3^k$, this is $k = \min\left(\lfloor v_2(2026!)/2 \rfloor,\ v_3(2026!)\right)$.

In binary, $2026 = 1024 + 512 + 256 + 128 + 64 + 32 + 8 + 2 = 11111101010_2$, which has eight 1s. By Legendre's Formula, $v_2(2026!) = 2026 - 8 = 2018$. For the prime 3: $v_3(2026!) = 675 + 225 + 75 + 25 + 8 + 2 = 1010$.

So $k = \min(1009, 1010) = 1009$. Notice that the bottleneck is the factor 4, not the 3; it is easy to get this wrong by analogy with base 10, where 5 is always the limiting prime.

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Problem 4

Suppose there is a solution other than the zero one. If $w = 0$ then $x^2 + y^2 + z^2 = 0$ forces $x = y = z = 0$, so in a nonzero solution $w \neq 0$. Choose a nonzero solution with $|w|$ as small as possible.

Recall that an odd square is 1 modulo 8 and an even square is 0 or 4 modulo 8. If w is odd, then $7w^2 \equiv 7 \pmod 8$. But a sum of three numbers from $\{0, 1, 4\}$ is one of $0, 1, 2, 3, 4, 5, 6, 8, 9, 12$, which are $0, 1, 2, 3, 4, 5, 6, 0, 1, 4$ modulo 8. It is never 7 modulo 8. This is a contradiction.

If w is even, then $7w^2 \equiv 0 \pmod 4$. Squares are 0 or 1 modulo 4, so $x^2 + y^2 + z^2$ is congruent modulo 4 to the number of odd terms among x, y, z, which is $0, 1, 2$ or 3. It must be 0, so x, y, z are all even. Then $(x/2, y/2, z/2, w/2)$ is an integer solution with $0 < |w/2| < |w|$, contradicting the minimality of $|w|$.

Both cases are impossible, so the only solution is $x = y = z = w = 0$. □

Problem 5

Answer: $n = 1$ and $n = 7$

Let $f(n) = n^4 + 2n^3 + 6n + 7$. The leading terms suggest comparing with $(n^2 + n)^2 = n^4 + 2n^3 + n^2$, and

$$f(n) - (n^2 + n)^2 = -n^2 + 6n + 7 = (7 - n)(n + 1).$$

Also $(n^2 + n - 1)^2 = n^4 + 2n^3 - n^2 - 2n + 1$, so $f(n) - (n^2 + n - 1)^2 = n^2 + 8n + 6 > 0$.

For $n > 7$ the first difference is negative and the second positive, so $(n^2 + n - 1)^2 < f(n) < (n^2 + n)^2$, and $f(n)$ lies between consecutive squares, so it is not a square. Hence $n \leq 7$.

For $n = 7$ the first difference is 0, so $f(7) = 56^2 = 3136$ is a square. For $n = 1, \ldots, 6$ the values are $16, 51, 160, 415, 912, 1771$. The first is 4^2, and the others are not squares, since $49 < 51 < 64$, $144 < 160 < 169$, $400 < 415 < 441$, $900 < 912 < 961$ and $1764 < 1771 < 1849$. So the answer is $n = 1$ and $n = 7$.

Problem 6

Answer: $(a, b) = (1, 1)$

The pair $(1, 1)$ works, since $7 - 3 = 4$. We show there are no others. If $b = 1$ then $7^a = 7$ and $a = 1$, so suppose $b \geq 2$.

Then $9 \mid 3^b$, so $7^a \equiv 4 \pmod 9$. The powers of 7 modulo 9 are $7, 4, 1$ and then repeat, since $7^3 = 343 \equiv 1$. Hence $a \equiv 2 \pmod 3$.

Now we want a modulus in which 3^b takes few values and in which knowing a modulo 3 restricts 7^a. A prime p with $\operatorname{ord}_p(3) = 3$ must divide $3^3 - 1 = 26$, and $p = 13$ works: $3^3 = 27 \equiv 1 \pmod{13}$, so $3^b \in \{1, 3, 9\}$ modulo 13. Modulo 13, $7^2 \equiv 10$, $7^3 \equiv 70 \equiv 5$ and $7^6 \equiv 25 \equiv -1$, so $\operatorname{ord}_{13}(7) = 12$, a multiple of 3. Since $a \equiv 2 \pmod 3$, a is congruent to $2, 5, 8$ or 11 modulo 12, and

$$7^2 \equiv 10, \quad 7^5 = 7^3 \cdot 7^2 \equiv 50 \equiv 11, \quad 7^8 = 7^6 \cdot 7^2 \equiv -10 \equiv 3, \quad 7^{11} = 7^6 \cdot 7^5 \equiv -11 \equiv 2.$$

So $3^b = 7^a - 4$ is congruent to one of $6, 7, 12, 11$ modulo 13, none of which lies in $\{1, 3, 9\}$. This contradiction shows $b \geq 2$ is impossible.

Therefore the only solution is $(a, b) = (1, 1)$. □

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